PROBLEM CL4

Cascade of Aerofoils: Downstream State

Original statement · CL4.pdf

Periodicity | Mass, momentum, energy | Perfect gas | Cascade

OFFICIAL PROBLEM STATEMENT

A planar uniform stream of air with pressure p_1, density ρ_1 and velocity $\mathbf{v}_1 = v_{1x}\mathbf{e}_x$ flows past an array of aerofoils with vertical separation L, exerting on each one a force (per unit length) given by $\mathbf{F} = F_x\mathbf{e}_x + F_y\mathbf{e}_y$. Assuming that the aerofoils are thermally insulated and neglecting the effect of gravity, write the equations that determine the values of the pressure p_2, density ρ_2 and velocity $\mathbf{v}_2 = v_{2x}\mathbf{e}_x + v_{2y}\mathbf{e}_y$ for the uniform stream found sufficiently far downstream.

Source: CL4.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46, force on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Steady planar flow of air through an infinite array of identical aerofoils, equally spaced by L in the y direction; the flow is periodic with period L. Far upstream (section 1) and far downstream (section 2) the flow is uniform.
- No gravity; the aerofoils are thermally insulated (adiabatic); viscous stresses and heat conduction are negligible in the uniform sections 1 and 2.
- Perfect gas with specific heat ratio γ: the specific enthalpy is $h = e + p/\rho = \dfrac{\gamma}{\gamma-1}\dfrac{p}{\rho}$. This is the assumption used in the official solution; the statement does not give γ.
- $\mathbf{F}$ is the force that the air exerts on each aerofoil, per unit length normal to the figure, so $[F_x/L]$ is a pressure. There are four unknowns, p_2, ρ_2, v_{2x}, v_{2y}, and therefore four equations are needed: mass, two components of momentum and energy.

Symbol	Meaning	SI unit
p_1, ρ_1, v_{1x}	upstream state (data)	Pa, kg/m^3, m/s
p_2, ρ_2, v_{2x}, v_{2y}	downstream state (unknowns)	Pa, kg/m^3, m/s
L	vertical separation (pitch) of the aerofoils	m
F_x, F_y	force on each aerofoil per unit length	N/m
γ	ratio of specific heats	1

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame fixed to the array, Cartesian basis $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, $\mathbf{e}_x$ along the incoming stream and $\mathbf{e}_y$ along the array.
- Fixed control volume (one period): Σ_1 (section far upstream, height L, $\mathbf{n} = -\mathbf{e}_x$), Σ_2 (section far downstream, height L, $\mathbf{n} = +\mathbf{e}_x$), the two stream surfaces Σ_3 (top) and Σ_4 (bottom), and the surface of the aerofoil Σ_5. $\mathbf{v}_c = 0$, so Eqs. 3.11, 3.36 and 3.46 apply with $\mathbf{v}_c = 0$.
- Periodicity: Σ_4 is Σ_3 translated by $-L\,\mathbf{e}_y$, and the outward normals are opposite at corresponding points ($\mathbf{n}_4 = -\mathbf{n}_3$). Therefore the integral over $\Sigma_3 \cup \Sigma_4$ of any periodic field times the normal vanishes: pressure force, viscous force, heat flux and work of viscous stresses.
- On the stream surfaces $\mathbf{v} \cdot \mathbf{n} = 0$ and on the aerofoil $\mathbf{v} = 0$ (no slip) with $\mathbf{q} \cdot \mathbf{n} = 0$ (insulated). The force of the aerofoil on the fluid is $-\mathbf{F}$ per unit length.

PHASE 3

Step-by-Step Derivation

3.1 Mass

Why this tool: Notes Eq. 3.11 with $\partial/\partial t = 0$. Only Σ_1 and Σ_2 are crossed by the flow, each of height L per unit length:

$$-\rho_1 v_{1x} L + \rho_2 v_{2x} L = 0 \implies \boxed{\rho_1\, v_{1x} = \rho_2\, v_{2x}}. \qquad \text{(E1)}$$

Denote by $\dot{m}' \equiv \rho_1 v_{1x} L$ the mass flow per unit length through one period.

3.2 Linear momentum, components x and y

Why this tool: the force $\mathbf{F}$ on the aerofoil enters only through the momentum balance (Notes Eq. 3.36). Left-hand side, using (E1) and $v_{1y} = 0$:

$$\int_{\Sigma_c} \rho\, \mathbf{v}\, (\mathbf{v}\cdot\mathbf{n})\, d\sigma = \left(-\rho_1 v_{1x}^2 L + \rho_2 v_{2x}^2 L\right) \mathbf{e}_x + \rho_2 v_{2y} v_{2x} L\, \mathbf{e}_y = \dot{m}' \left[(v_{2x} - v_{1x})\, \mathbf{e}_x + v_{2y}\, \mathbf{e}_y\right].$$

Right-hand side: on Σ_1, $-p_1(-\mathbf{e}_x)L = p_1 L\, \mathbf{e}_x$; on Σ_2, $-p_2 L\, \mathbf{e}_x$; the contributions of Σ_3 and Σ_4 cancel by periodicity; viscous stresses are negligible on Σ_1 and Σ_2; on Σ_5 the force on the fluid is $-\mathbf{F}$. Component by component:

$$\dot{m}'\,(v_{2x} - v_{1x}) = (p_1 - p_2)\, L - F_x \implies \boxed{\frac{F_x}{L} = p_1 - p_2 + \rho_1 v_{1x}\,(v_{1x} - v_{2x})} \quad \text{(E2)}$$

$$\dot{m}'\, v_{2y} = -F_y \implies \boxed{\frac{F_y}{L} = -\rho_1\, v_{1x}\, v_{2y}} \quad \text{(E3)}$$

3.3 Total energy

Why this tool: p_2 and ρ_2 are linked by the thermal state of the gas, which only the energy balance (Notes Eq. 3.46) provides. Steady, no gravity, no volumetric heat: the convective flux of $e + v^2/2$ plus the pressure work $p\,\mathbf{v}\cdot\mathbf{n}$ combine into the flux of $\rho\left(h + v^2/2\right)$.

On Σ_3 and Σ_4: $\mathbf{v}\cdot\mathbf{n} = 0$, and conduction and viscous work cancel by periodicity. On Σ_5: $\mathbf{v} = 0$ (no work) and $\mathbf{q}\cdot\mathbf{n} = 0$. On Σ_1 and Σ_2: viscous work and conduction are negligible. What remains is

$$\int_{\Sigma_1+\Sigma_2} \rho \left(h + \frac{v^2}{2}\right) (\mathbf{v}\cdot\mathbf{n})\, d\sigma = 0 \implies \dot{m}' \left[\left(h_2 + \frac{v_{2x}^2 + v_{2y}^2}{2}\right) - \left(h_1 + \frac{v_{1x}^2}{2}\right)\right] = 0,$$

$$\boxed{\frac{\gamma}{\gamma-1}\frac{p_2}{\rho_2}+\frac{v_{2x}^2+v_{2y}^2}{2}=\frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1}+\frac{v_{1x}^2}{2}} \qquad \text{(E4)}$$

Equations (E1) to (E4) are four equations for the four unknowns p_2, ρ_2, v_{2x} and v_{2y}, which answers the question of the statement.

PHASE 4

Interpretation, Limits and Dimensional Check

- Equation (E3) says that the transverse force on an aerofoil (lift) equals minus the rate of change of y momentum of the stream: the flow is deflected by v_{2y} in the direction opposite to the force on the body. Equation (E2) splits the streamwise force into a pressure drop across the array and a momentum change.
- Trivial limit $\mathbf{F}=0$: (E1) to (E4) are satisfied by $\rho_2=\rho_1, p_2=p_1, \mathbf{v}_2=\mathbf{v}_1$ (no body, no change).
- Incompressible limit ($\rho_2=\rho_1$): (E1) gives $v_{2x}=v_{1x}$ and (E2) reduces to $p_1-p_2=F_x/L$.
- Dimensions (SI): $[F_x/L]=(\mathrm{N/m})/\mathrm{m}=\mathrm{Pa}$ and $[\rho_1 v_{1x}^2]=\mathrm{Pa}$; $[\rho_1 v_{1x}\,v_{2y}\,L]=(\mathrm{kg/m^3})(\mathrm{m/s})(\mathrm{m/s})\,\mathrm{m}=\mathrm{N/m}$; $[\gamma p/((\gamma-1)\rho)]=\mathrm{m^2/s^2}=[v^2]$.

Extension (beyond the official solution): closed solution of the four equations

With $m\equiv\rho_1 v_{1x}$, (E1) gives $\rho_2=m/v_{2x}$, (E3) gives $v_{2y}=-F_y/(mL)$ and (E2) gives $p_2=P-m\,v_{2x}$ with $P\equiv p_1+m\,v_{1x}-F_x/L$. Substituting into (E4), with $E\equiv\frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1}+\frac{v_{1x}^2}{2}$, and using that the coefficient of v_{2x}^2 is $\frac12-\frac{\gamma}{\gamma-1}=-\frac{\gamma+1}{2(\gamma-1)}$, one obtains the quadratic $(\gamma+1)\,v_{2x}^2-2\gamma\,\frac{P}{m}\,v_{2x}+2(\gamma-1)\left(E-\frac{v_{2y}^2}{2}\right)=0$. For $\mathbf{F}=0$ one root is $v_{2x}=v_{1x}$; the root continuous with it is the physical one (the other is the supersonic-subsonic conjugate state, as in a normal shock). Numerical check: $p_1=101325$ Pa, $\rho_1=1.2$ kg/m^3, $v_{1x}=100$ m/s, $L=0.2$ m, $F_x=30$ N/m, $F_y=-80$ N/m, $\gamma=1.4$ give $v_{2x}=100.16$ m/s, $v_{2y}=3.3333$ m/s, $\rho_2=1.19808$ kg/m^3, $p_2=101155.8$ Pa, with residuals of (E1) to (E4) below 10^{-9} in SI units; the other root gives $p_2=-6843.3$ Pa <0 and is discarded.